

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Scenario-Based

1. A newly established engineering college is assigned the IP block **192.168.100.0/24** to design its internal network. The network administrator has already assigned subnet masks for each department as follows: Computer Science $\rightarrow$ /25, Electronics $\rightarrow$ /26, and Mechanical $\rightarrow$ /27. Using the given subnet masks, apply subnetting techniques to determine the network address and broadcast address for each department. Identify the valid range of host IP addresses for each subnet. Calculate the number of usable hosts available in each subnet. Finally, determine the remaining unused IP address range within the given network block.

2. A software company is provided with the IP block **10.10.0.0/16** for its internal network. The network engineer has already assigned subnet masks for each team as follows: Development $\rightarrow$ /24, Testing $\rightarrow$ /25, HR $\rightarrow$ /26, and Management $\rightarrow$ /27. Using the given subnet masks, apply subnetting techniques to determine the network address and broadcast address for each team. Identify the valid range of host IP addresses for each subnet. Calculate the number of usable hosts available in each subnet. Finally, determine the remaining unused IP address range within the given IP block.

3. An organization is assigned the IPv6 address **2001:0db8:abcd:0000:0000:0000:0000:0000/64** for its network configuration. The network administrator requires both compressed and expanded forms of the given IPv6 address. Apply IPv6 addressing rules to convert the given address into its compressed form and then expand it back to its full notation. Using the prefix length /64, determine the network prefix of the given address. Calculate the total number of possible host addresses available within this network.

4. A company has interconnected three branch offices using routers with the subnets **192.168.1.0/24**, **192.168.2.0/24**, and **192.168.3.0/24**. A data packet arrives at the central router with the destination IP address **192.168.2.45**. Using the given subnet information, apply routing techniques to determine the appropriate network to which the destination IP belongs. Identify the matching route entry based on the destination address. Determine the next-hop network for forwarding the packet. Also, identify the default route that can be used for destinations not present in the routing table.

5. A small office network consists of two LANs connected through a router. The first LAN uses the subnet **192.168.10.0/26**, and the second LAN uses **192.168.10.64/26**. The network administrator assigns IP addresses to the router interfaces as **192.168.10.1** for the first LAN and **192.168.10.65** for the second LAN. Using the given configuration, apply subnetting techniques to determine the valid host IP range for each LAN. Identify suitable IP addresses that can be assigned to computers in both LANs. Determine the network and broadcast address for each subnet. Also, identify the gateway address used by hosts in each LAN for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand